

gSender Android performance report

Build 53 | 21 September 2026

Build 53 greatly reduces delayed job-status updates and completes the seven-case suite on all three tablets. Lenovo and K50 animation improved; K90 relief animation regressed. The changes below cover the Android project cumulatively against official gSender 1.6.4. Measured before-and-after results compare Android Builds 52 and 53, not stock desktop gSender.

Measured changes from Build 52 to Build 53

Tablet	Relief completion display delay s	Relief frame gap p95 ms	40 MiB load and preview s
Lenovo TB-8506F	69.0 → 0.55	117 → 50	Not reached → 42.6
Rebecca K50	23.8 → 0.55	116 → 50–66	28.8 → 17.3
Rebecca K90	43.1 → 0.57	117 → 150	30.7 → 17.7

Lower values are better. Frame gaps measure animation callbacks during streaming, not presented FPS or finger latency. K90's 150 ms result is a regression. Lenovo's prior suite timed out loading 20 MiB; Build 53 loaded it in 22.1 s.

Cumulative improvements relative to live gSender

Android execution and connections. The desktop application is hosted in Android WebView with its existing backend and native USB bridge. The port moved from Node.js Mobile to a source-built Node 24.21.0 runtime, added USB recovery and SLB/Lite autoconnection, and retained scoped dev pendant/jogging code with Android touch and sensor fixes. These are portability and reliability changes, not measured desktop speed gains.

Less routine interface work. Toolpath geometry is retained while the position marker moves. Progress reuses line totals or counts incrementally. Hidden knob polling and disabled log formatting are avoided; release code and resources are optimized. A component test reduced 30 position-triggered geometry rebuilds to zero.

Lower job-storage overhead. The sender uses an 8-byte offset index per nonblank command instead of retaining individual line strings. A host-only 20 MiB microbenchmark reduced extra line storage from 26.1 to 4.4 MiB. RAM-aware budgeting replaces the port's early fixed heap limits; this ARM32 APK allows up to 1,536 MiB of V8 old-space, not total app memory.

Bounded and cached previews. Worker-rendered bitmaps replace large pendant SVG paths, with a 1,600-pixel edge limit and bounded drawing batches. Reused surfaces and transferable images avoid repeated encoding and redraws. Stale workers and images are released; original CNC commands retain their precision.

Fresher status with less console traffic. Build 53 batches display copies of serial lines, bounds console history, pauses hidden-console rendering, and coalesces superseded position/progress snapshots. GrblHAL status polling changes from 250 to 100 ms. Alarms and state transitions retain ordering; command transmission and acknowledgements remain intact.

Repeatable diagnostics. Tools now includes offline job simulation, pan and 2× zoom workloads, checksum verification, memory/heartbeat logging, preview-stage errors and JSON export. These additions make regressions measurable on each tablet.

Device statistics for Build 53

Statistic	Lenovo TB-8506F	K50	K90
Android version	11	15	16
Android reported RAM GiB	1.77	3.70	8.00
Minimum free system RAM MiB	335	1,313	4,768
Peak host and Node PSS MiB	256	396	584
Maximum heartbeat response age ms	306	307	71
Full suite duration minutes	20.33	19.12	19.12

RAM is reported by the onboard Android API, not a hardware specification. Host PSS excludes the isolated WebView renderer. No sampled low-memory flags or preview failures occurred. Lenovo's 335 MiB minimum was above the 256 MiB test stop threshold but leaves limited headroom.

Loading and preview completion in Build 53

Job	Lenovo seconds	K50 seconds	K90 seconds
Contour 128 KiB	0.96	0.72	1.02
Arcs 1 MiB	10.13	4.78	5.17
Real mountain relief 4.77 MiB	9.40	4.01	3.96
Dense relief 5 MiB	6.47	3.17	3.09
Dense relief 20 MiB	22.10	9.21	8.98
Dense relief 30 MiB	32.83	13.38	13.61
Dense relief 40 MiB	42.59	17.29	17.68

Test coverage and development evidence

Each tablet verified 328,139 commands across three complete jobs with matching ordered checksums and zero receive overflows. Full-job rates were 493.6–500.6 commands/s against a 500 commands/s target. The 5–40 MiB cases used roughly 65–80-second streaming samples. All cases included idle and streaming viewport activity in portrait orientation.

Build 45 exposed K90 graphics allocation failures at 30–40 MiB; bounded previews in Builds 46–47 enabled those sampled jobs to load and stream. Build 48's GPU-raster experiment caused a Lenovo unresponsive-app event and was rejected. Build 51 added bitmap caching while retaining worker software rasterization and a hardware-accelerated WebView. Builds 52–53 added onboard tests and display fixes.

Remaining performance work

Prioritize K90 rendering: its viewport matched the baseline and heartbeat responses stayed timely, but the cause is unconfirmed. Repeat and profile before ranking hardware. Tablet files above 40 MiB, long endurance runs, physical USB/CNC throughput and ARM64 delivery remain unqualified. A host-only 100 MiB storage test does not establish tablet support at that size.

Build 53 passed 322 automated tests, frontend builds, Android release assembly/lint and APK/runtime checks. No unrelated UI-task changes were included. Findings use one full run per tablet; memory samples can miss peaks, and the simulator shares tablet resources.

Sources: [Official gSender 1.6.4 release](#); [MAXMAKE relief project](#). Local evidence: Build 52/53 tablet JSON exports, Build 22/46/47/51 notes, MEMORY-PERFORMANCE.md and upstream.json. Relief fixture replaces tool-change stops for unattended simulation; it is not a machine-qualified cutting file.